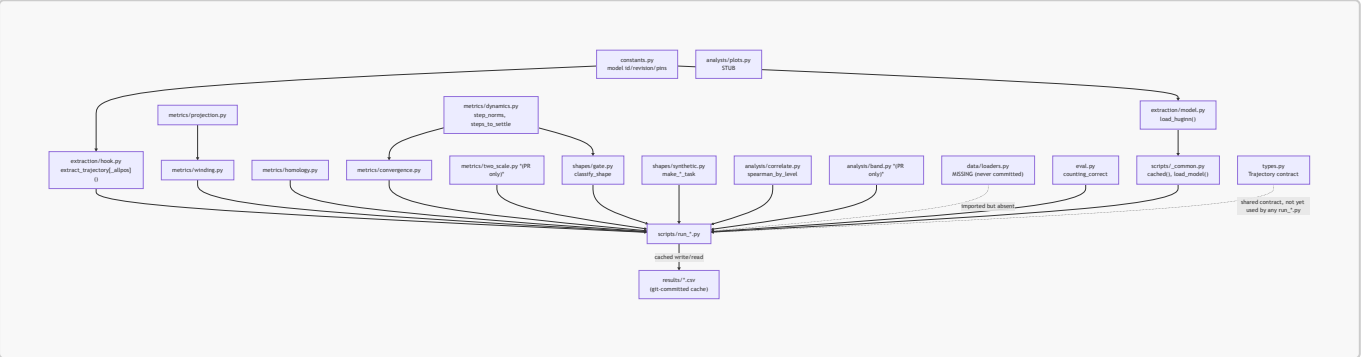
Code infrastructure — architecture, conventions, cookbook

Distinct from the other three docs in here: `research_log.md` is the discussion trace, `guide.md` is "how do I set up and reproduce this," `claims_ledger.md` is "is this specific number true." This one is the architecture — module dependency structure, the team conventions baked into the code, and a cookbook for extending it. All figures below were extracted with grep/git, not estimated.

1. Module dependency graph



Three concentric layers, by GPU/model dependency:

1. **Model-bound** (needs CUDA + the `model` extra): `extraction/model.py`, `extraction/hook.py`, `eval.py`. Nothing else touches `torch` or `transformers` directly — imports of these are always deferred (`from traj_geom.extraction.hook import extract_trajectory` *inside* `compute()` functions, never at module top of a metric/shape/analysis file) so that importing the rest of the package never requires a GPU or even `torch` installed.
2. **Pure numpy/scipy/sklearn, model-free** (everything in `metrics/`, `shapes/`, `analysis/`, `types.py`): operates on a plain `[T, hidden_dim]` (or `[T, seq_len, hidden_dim]`) array. This is why it's the fully-tested layer — synthetic ground-truth arrays (circles, spirals, lines) exercise it without ever touching Huginn.
3. **Glue** (`scripts/run_*.py`, `scripts/_common.py`): the only layer that imports across both of the above, and the only layer that talks to `results/*.csv`.

`types.py::Trajectory` is worth flagging explicitly: it's described in its own docstring as "the ONE fully implemented module... every teammate codes against `Trajectory`," but **no `run_*.py` script actually constructs or consumes a `Trajectory` object** — every script passes bare numpy arrays around instead. The dataclass (with its `.save()`/`.load()` npz round-trip, tested in `test_types.py`) exists and works, but the shared-contract vision in its docstring isn't actually how the codebase evolved. Not a bug, just a gap between an early design doc and what got built — worth knowing before assuming trajectories flow through the codebase as `Trajectory` objects anywhere except that one test.

2. Environment & dependency infra

Extra	Adds	Needed for
(base)	numpy, scipy, scikit-learn, matplotlib, pandas, tqdm, pyyaml	All CPU-only metrics/shapes/analysis code, and reading cached `results/*.csv`
`model`	torch, transformers==4.53.3, accelerate, datasets	Loading Huginn and running `extract_trajectory*` — needs a GPU in practice (`.to("cuda")` is hardcoded, see `guide.md` §3)
`tda`	ripser, persim, scipy	`metrics/homology.py` / `run_homology.py` only
`dev`	pytest, ruff	Running the test suite / linting — **not installed by plain `uv sync`**, see `guide.md` §5

`transformers` is pinned to exactly `4.53.3` (working window documented as 4.50–4.53) with a specific model revision hash (`bb6621b65e90b6a4b9b29ef88dc83866d450470c`) — both load-bearing, not cosmetic. `constants.py`'s docstring spells out why, verbatim from debugging: `4.49-` lacks a `device` arg the custom Huginn code needs, `4.54+` makes `key_cache` read-only (breaks the custom modeling code), `5.x` changes tied-weights loading. If you ever bump `transformers` here, expect the model load itself to break, not a downstream metric.

Python is pinned to `>=3.11,<3.12` (`.python-version` says `3.11` exactly) project-wide, no stated reason beyond "this is what it was built/tested on."

No CI (no `.github/` directory at all, on any branch), no Dockerfile, no cloud/notebook infra checked in beyond the two Jupyter notebooks themselves. GPU compute is manual, on Kaggle (`guide.md` §3) — reproducibility for anything touching the model depends entirely on someone with Kaggle access running the relevant `scripts/run_*.py` by hand and committing the resulting CSV.

3. The OWNER convention — and its actual authorship

Every `src/traj_geom/**/*.py` and `scripts/run_*.py` file on `main` opens with an `OWNER:` tag (plus `STATUS:`, `TASK:`, `I/O:`). Exact counts, grepped directly:

- **Extraction+Winding** — 13 files: model loading, the hook, winding, projection, convergence, homology, dynamics, `run_contrast.py`, `run_homology.py`, `run_convergence.py`. The model-facing / core-dynamics lane.
- **Data+Analysis** — 12 files: `correlate.py`, `plots.py` (stub), `eval.py`, and the majority of experiment scripts (`run_pararule`, `run_dissociation`, `run_dissociation_multiinit`, `run_counting`, `run_switch`, `run_maxtask`, `run_accuracy`, `run_mvp`). The stats + experiment-orchestration lane — also the biggest one.
- **Shapes+Gate** — 5 files: `gate.py`, `synthetic.py`, `run_forceloop.py`, `run_phase.py`. The regime-classification + task-design lane.

Important caveat, checked against actual git blame, not assumed: on `main`, every one of these files was written by a single person (`Shtirmann` — and `Alexander Shiyanov`, the other name in `main`'s history, is the same person: their commits use GitHub's `@users.noreply.github.com` address which embeds the `Shtirmann` handle, i.e. that's the same account committing via the GitHub web UI for two doc-only commits vs the local git client for everything else). **The three OWNER lanes are the team's intended division of labor, not a record of who actually built each part** — nobody but Shtirmann has

institutional first-hand knowledge of any of this code yet, role tags notwithstanding. `jack`'s only contribution (the two-scale feature branch) doesn't use the OWNER convention consistently — only the PR's new `analysis/band.py` does (tagged Data+Analysis), matching `main`'s style; the older `metrics/two_scale.py` functions don't have OWNER headers at all. If you're picking a lane to start contributing in, all three are equally "unclaimed" in practice.

4. The caching contract — the load-bearing pattern

`scripts/_common.py::cached(name, compute_fn)` is small (20 lines) but it's the single piece of infrastructure that makes this whole project usable by more than one person with GPU access:

```
def cached(name, compute_fn):
    path = RESULTS_DIR / name
    if path.exists():
        return pd.read_csv(path)        # <- no GPU, no model import
triggered
    df = compute_fn()                    # <- only reached on a cache miss
    df.to_csv(path, index=False)
    return df
```

Every `run_*.py`'s `compute()` does its `torch`/`transformers` imports *inside* the function body specifically so that hitting the cache never imports them. This is why `uv run python -m scripts.run_pararule` (once the missing loader is fixed, see `guide.md` §5) works with zero GPU as long as `results/pararule.csv` exists — and why deleting a CSV to force a re-run is a real GPU-requiring action, not a formality.

Practical implication: **the `results/*.csv` files are the actual dataset of this project.** Editing a metric's formula after a result was cached does **not** retroactively change anything — the stale CSV is silently reused until someone deletes it and reruns with GPU access. There's no hash/version check tying a cached CSV to the code that produced it. If you change `metrics/winding.py`, for instance, every `results/*winding*` number is now stale until manually invalidated.

5. Testing infra (see `claims_ledger.md` D1/D2/C9 for current pass/fail state)

Tests exist only for the model-free layer (§1, layer 2): `test_convergence`, `test_correlate`, `test_gate_synthetic`, `test_homology` (skipped without `ripser`), `test_types`, `test_winding_synthetic` on `main`; the PR branch adds `test_two_scale.py` (currently fails to collect — the loaders bug) and a throwaway `test_simple.py`. Nothing tests `extraction/hook.py` or `extraction/model.py` — untestable without a GPU and 3.5B weights, so the gotchas that would normally be caught by a test live instead as prose warnings in `constants.py` and the module docstrings ("hard-won," "keep verbatim"). Read those before touching extraction code; they're the closest thing this repo has to regression tests for the model-loading path.

6. Cookbook

Add a new synthetic reasoning task: write a `make_<name>_task(n_ops, seed) -> dict` in `shapes/synthetic.py` (follow the existing `make_counting_task`/`make_switch_task` shape:

`{"prompt", "answer"}` or a `{"track", "local"}` pair if you want a length-matched dissociation control for free). Reuse `extraction.hook.extract_trajectory`, `metrics.dynamics`, `metrics.winding` in a new `scripts/run_<name>.py` modeled on `run_switch.py` (it's the shortest complete example, 57 lines) — wrap the GPU part in `compute()`, call it through `cached()`, report with `fmt_by_level` from `analysis/correlate.py`.

Add a new metric: put it in `metrics/<name>.py` as a pure function on a `[T, hidden_dim]` array (no torch/model imports), give it an OWNER header, and write a test against a synthetic array with known ground truth (a circle for anything winding/loop-related, a damping spiral for anything convergence-related — both helpers already exist inline in `tests/test_convergence.py` and `tests/test_winding_synthetic.py`, worth factoring out into a shared `tests/synthetic_paths.py` if you add a third consumer).

Add a new experiment script: copy the shape of `run_switch.py` or `run_maxtask.py` (both ~55 lines, single task, single metric pass) rather than the larger multi-stage ones (`run_dissociation_multiinit.py`, `run_two_scale_depth.py`) unless you specifically need length-matching or multi-seed robustness — those patterns exist in `analysis/band.py` and `run_dissociation_multiinit.py` if you do.

Debugging a Huginn load/hook problem: read `constants.py`'s docstring gotchas first, in full, before touching `extraction/`. In order of what's bitten someone already: wrong `transformers` version, `num_steps` passed as a tensor instead of a plain int, a hook left registered by a crashed previous run doubling the captures (always `_forward_hooks.clear()` + try/finally), calling `.generate()` instead of `.forward()` for trajectory work.

Running on Kaggle (inferred from `notebooks/01_mvp.ipynb`, not written down anywhere else): `!pip install -q "transformers==4.53.3" datasets accelerate`, set `CACHE = "/kaggle/working"` (persists as the notebook's output), then the notebook cells are effectively the same `compute()` / `cached()` pattern as the scripts, just inline. Porting a notebook cell into `src/traj_geom` after it works is the established workflow (`guide.md` §2) — every module's `STATUS:` line says which notebook cell it came from.

7. Known infra bugs

Not repeated here in full — see `guide.md` §5 for the verified list (`data/loaders.py` never committed on any branch due to an overbroad `.gitignore:2 data/` rule; the README's own `uv sync ; uv run pytest` reproduce command is broken; `analysis/plots.py` is an unimplemented stub).